Huawei Access Node Optical Transport Failure Resolution SOP (JAY-ACC-77)

Node: JAY-ACC-77 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6897

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node (JAY-ACC-77) at Jayanagar Metro Site is a critical failure domain that can occur due to the inherent limitations of optical transmission technology. In this scenario, the failure is likely caused by the degradation of the SFP/XFP transceiver, which can be attributed to the cumulative effect of temperature fluctuations, vibration, and optical signal degradation over time. The Huawei Access Node, being a key component in the optical transport network, plays a crucial role in forwarding optical signals between nodes. The failure of this node can have a cascading effect on the entire network, resulting in service disruptions and data loss. To mitigate this risk, it is essential to promptly identify and resolve the root cause of the failure, which can be achieved through a combination of physical and logical checks, including SFP/XFP inspection, fiber optic cable verification, and optical signal analysis.

Pre-requisite Checks

- Check: Verify the redundant link status on the Access Node (JAY-ACC-77) to ensure that it is functioning properly., Description: This check is crucial to ensure that the network remains operational even in the event of a primary link failure.

- Check: Dispatch a remote hand to the Jayanagar Metro Site to physically inspect the SFP/XFP transceiver and verify the integrity of the fiber optic cable., Description: This physical inspection is essential to identify any signs of degradation or damage to the SFP/XFP transceiver or the fiber optic cable.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber optic cable damage or degradation	Medium
Optical signal loss due to incorrect transceiver settings	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver info transceiver-index 1, Expected Output: Transceiver Index: 1, Type: SFP, Status: Degraded, Description: This command is used to display the transceiver information, including the type, status, and index. The expected output indicates that the SFP transceiver is degraded, which could be the root cause of the failure.
```

```
Command: display optical signal info optical-index 1, Expected Output: Optical Index: 1, Power: -10 dBm, Signal Quality: Poor, Description: This command is used to display the optical signal information, including the power and signal quality. The expected output indicates that the optical signal is poor, which could be due to fiber optic cable damage or degradation.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceiver, Command: replace transceiver transceiver-index 1, Description: This step involves replacing the degraded SFP/XFP transceiver with a new one to ensure that the optical signal is transmitted correctly.

Step 2: Step: Verify the fiber optic cable integrity, Command: display fiber info fiber-index 1, Description: This step involves verifying the integrity of the fiber optic cable to ensure that it is not damaged or degraded.

Post-Check Verification

```
Command: display transceiver info transceiver-index 1, Expected Output: Transceiver Index: 1, Type: SFP, Status: Normal, Description: This command is used to verify that the transceiver is functioning properly after replacement.

Command: display optical signal info optical-index 1, Expected Output: Optical Index: 1, Power: 0 dBm, Signal Quality: Good, Description: This command is used to verify that the optical signal is transmitted correctly after replacing the transceiver and verifying the fiber optic cable integrity.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore transceiver transceiver-index 1, Description: This procedure involves restoring the original SFP/XFP transceiver in case the replacement fails to resolve the issue.